



MAHARSHI DAYANAND UNIVERSITY, ROHTAK

PROJECT REPORT

ON

**Machine Learning Pipeline for Biomedical Text
Analytics Using Orange**

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ABSTRACT

Biomedical literature is growing at an unprecedented pace, making manual analysis impractical. This project presents a complete machine-learning pipeline for text analytics using Orange Data Mining. The workflow integrates text preprocessing, TF-IDF vectorization, Principal Component Analysis (PCA), cosine distance computation, hierarchical clustering, supervised classification, and prediction of unseen biomedical abstracts. PCA successfully reduced the dimensionality of high-dimensional textual data, while hierarchical clustering revealed distinct semantic groupings among biomedical domains such as COVID-19, PEDOT electrodes, electroactive polymers, deep learning, human connectome, neuroprosthetics, and brain-machine interfaces. Among all classifiers tested, Logistic Regression achieved the highest accuracy (83.8%) and F1-score (0.845). Predictive evaluation on new unseen abstracts confirmed the model's generalization ability. The project demonstrates the effectiveness of NLP-ML workflows in biomedical literature categorization and knowledge discovery.

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1. INTRODUCTION

The exponential growth of biomedical research publications presents significant challenges for manual literature curation, organization, and interpretation. Every year, thousands of new articles are published across domains such as neuroscience, biomedical engineering, computational biology, infectious diseases, and medical imaging. With the increasing amount of scientific knowledge being generated, traditional manual reading and classification of research articles has become impractical, time-consuming, and error-prone. There is a pressing need for automated systems capable of processing large textual datasets, identifying patterns, discovering hidden relationships, and categorizing content in a structured manner.

Text mining and Natural Language Processing (NLP) have emerged as powerful tools to extract actionable information from unstructured biomedical text. By combining computational linguistics, statistics, machine learning, and domain-specific knowledge, NLP enables scalable analysis of scientific literature. It supports tasks such as document classification, clustering, topic discovery, literature triaging, drug-disease association extraction, and biomarker discovery. In modern bioinformatics, text mining has become an essential component for accelerating research and decision-making.

Machine learning plays a critical role in improving the accuracy and efficiency of text analysis. Techniques such as supervised classification, unsupervised clustering, dimensionality reduction, and similarity computation are widely used for large-scale knowledge extraction. Biomedical abstracts in particular contain rich domain-specific terminology, making them suitable for machine-learning-based categorization. TF-IDF (Term Frequency-Inverse Document Frequency) is a widely used vectorization technique that transforms text into numerical representations, enabling downstream ML algorithms to operate effectively.

Orange Data Mining, a visual machine-learning platform, was used in this project to design a complete text-mining pipeline without coding. It provides integrated tools for preprocessing, feature extraction, dimensionality reduction, clustering, classification, and model evaluation, making it suitable for biomedical text analytics.

This project presents a comprehensive machine learning pipeline for biomedical text analytics using Orange. The workflow processes biomedical research abstracts from eight distinct scientific domains: deep learning, COVID-19, human connectome, virtual reality, electroactive polymers, PEDOT electrodes, neuroprosthetics, and brain-machine interfaces. These topics were selected because they represent emerging research areas across computational biology, neuroscience, materials science, and medical engineering.

The project incorporates several key components:

1. **Text preprocessing** to clean and normalize the data
2. **TF-IDF vectorization** to convert text to numerical form
3. **Principal Component Analysis (PCA)** to reduce dimensionality and reveal semantic structure
4. **Cosine distance computation** for similarity measurement
5. **Hierarchical clustering** to identify natural groupings within the dataset
6. **Evaluation of multiple machine-learning classifiers**

7. Prediction of categories for unseen biomedical abstracts

By integrating NLP, unsupervised learning, and supervised machine learning, this project demonstrates how automated systems can classify and analyze biomedical literature with high accuracy. The results highlight strong performance of Logistic Regression and provide clear insights into the semantic relationships between different biomedical topics.

This work contributes to the field by presenting a reproducible pipeline for scientific text classification and by demonstrating the usefulness of Orange as a visual tool for bioinformatics-related machine learning tasks. The approach can be extended to large-scale literature databases, systematic reviews, and biomedical knowledge extraction systems.

2. OBJECTIVES

1. To preprocess biomedical research abstracts using NLP techniques.
2. To convert textual data into numerical TF-IDF features.
3. To reduce feature dimensionality using Principal Component Analysis (PCA).
4. To compute cosine distances for similarity measurement.
5. To apply hierarchical clustering for unsupervised grouping of abstracts.
6. To evaluate multiple machine learning classifiers.
7. To predict research categories for unseen biomedical abstracts.

3. LITERATURE REVIEW

Biomedical text mining has become an essential component of modern bioinformatics and computational biology. Large databases such as PubMed, BioRxiv, and Medline contain millions of abstracts covering diverse scientific fields, ranging from infectious diseases and neuroscience to biomedical engineering and computational modeling. Extracting meaningful insights from this vast volume of information requires automated text-mining tools and machine-learning techniques capable of processing unstructured data efficiently.

Natural Language Processing (NLP)

NLP enables computers to interpret and manipulate human language. In biomedical applications, NLP is used for disease terminology extraction, gene-protein relation mining, clinical diagnosis classification, and literature triage. Techniques such as tokenization, lemmatization, and stop-word removal serve as foundational steps for reducing linguistic variability and preparing text for vectorization.

Text Vectorization Techniques

Among various text representation methods, TF-IDF (Term Frequency-Inverse Document Frequency) remains one of the most widely used due to its interpretability and computational efficiency. TF-IDF assigns higher importance to domain-specific terms while down-weighting frequent but uninformative words. This proves especially useful for biomedical literature, where technical terms often carry significant meaning.

Dimensionality Reduction in Text Mining

Biomedical text produces extremely high-dimensional vectors, often exceeding thousands of features. Principal Component Analysis (PCA) is frequently employed to reduce dimensionality while retaining

underlying semantic structure. PCA transforms correlated TF–IDF features into orthogonal principal components, making the dataset suitable for visualization, clustering, and classification.

Clustering Techniques for Biomedical Text

Hierarchical clustering is commonly used to explore structural relationships in biomedical literature. Unlike K-means, hierarchical clustering does not require specifying the number of clusters beforehand. It visualizes nested grouping via a dendrogram, making it useful for understanding similarities among diverse biomedical topics such as COVID-19, neuroprosthetics, and deep learning.

Machine Learning for Text Classification

Supervised models such as Logistic Regression, Naive Bayes, Random Forest, and Neural Networks are widely applied in biomedical NLP tasks.

- Logistic Regression excels with linearly separable high-dimensional data like TF–IDF vectors.
- Naive Bayes, although computationally efficient, often struggles due to independence assumptions.
- Random Forest performs well on nonlinear structures but may be less effective in sparse text data.
- Neural Networks capture complex patterns but require careful tuning to avoid overfitting.

Recent studies highlight that linear models often outperform deep models when using PCA-reduced TF–IDF vectors, which aligns with the findings of this project.

Applications in Biomedical Research

Biomedical text classification supports tasks such as:

- automated categorization of scientific papers
- literature recommendation systems
- drug–disease relation identification
- clinical decision support
- systematic review acceleration

The integration of these techniques in Orange provides an educational yet powerful platform for developing and validating NLP workflows without coding barriers.

This literature establishes the foundation for the methodology used in this project, demonstrating the importance of text-mining pipelines in modern biomedical informatics.

4. MATERIALS AND METHODS

The workflow implemented in this project integrates Natural Language Processing (NLP), text vectorization, dimensionality reduction, clustering, and machine-learning classification using Orange Data Mining. The overall methodology follows standard practices in biomedical text analytics, as supported by prior literature.

4.1 Dataset Description

The dataset consists of **400 biomedical research abstracts**, distributed evenly across **eight scientific categories**, including deep learning, COVID-19, human connectome, virtual reality, electroactive polymers, PEDOT electrodes, neuroprosthetics, and brain–machine interfaces. Each category contains 50 curated abstracts designed to represent distinct biomedical domains. An additional **10 unseen unlabelled abstracts** were used for external evaluation of the machine-learning models.

Biomedical abstracts are widely used in text-mining research due to their structured vocabulary and domain-rich terminology (Hirschberg & Manning, 2015).

4.2 Software Environment: Orange Data Mining

Orange Data Mining (version 3.x) was used as the primary environment for building and executing the machine-learning workflow in this study. Orange is an **open-source, visual data-mining and machine-learning platform** that enables the construction of analytical pipelines through a simple drag-and-drop interface. Unlike traditional programming-based approaches that require coding expertise in Python or R, Orange provides an intuitive workflow system where each analytical step (preprocessing, feature extraction, dimensionality reduction, clustering, classification, evaluation) is represented as a modular widget.

Orange was chosen for this project for several key reasons:

1. No-Code, Highly Visual Workflow
2. Built-In Text Mining Add-On
3. Seamless Integration of ML + NLP
5. Real-Time Visualization
6. Suitable for Educational and Research Use
7. Quick Experimentation and Model Comparison

4.3 Workflow Design in Orange

The data-analysis pipeline was implemented in **Orange Data Mining**. The workflow included the following modules:

1. Preprocess Text
2. Select Columns
3. Bag of Words (TF–IDF)
4. Principal Component Analysis (PCA)
5. Distances (Cosine)
6. Hierarchical Clustering
7. PCA Scatter Plot
8. Classification Models
9. Test & Score
10. Predictions

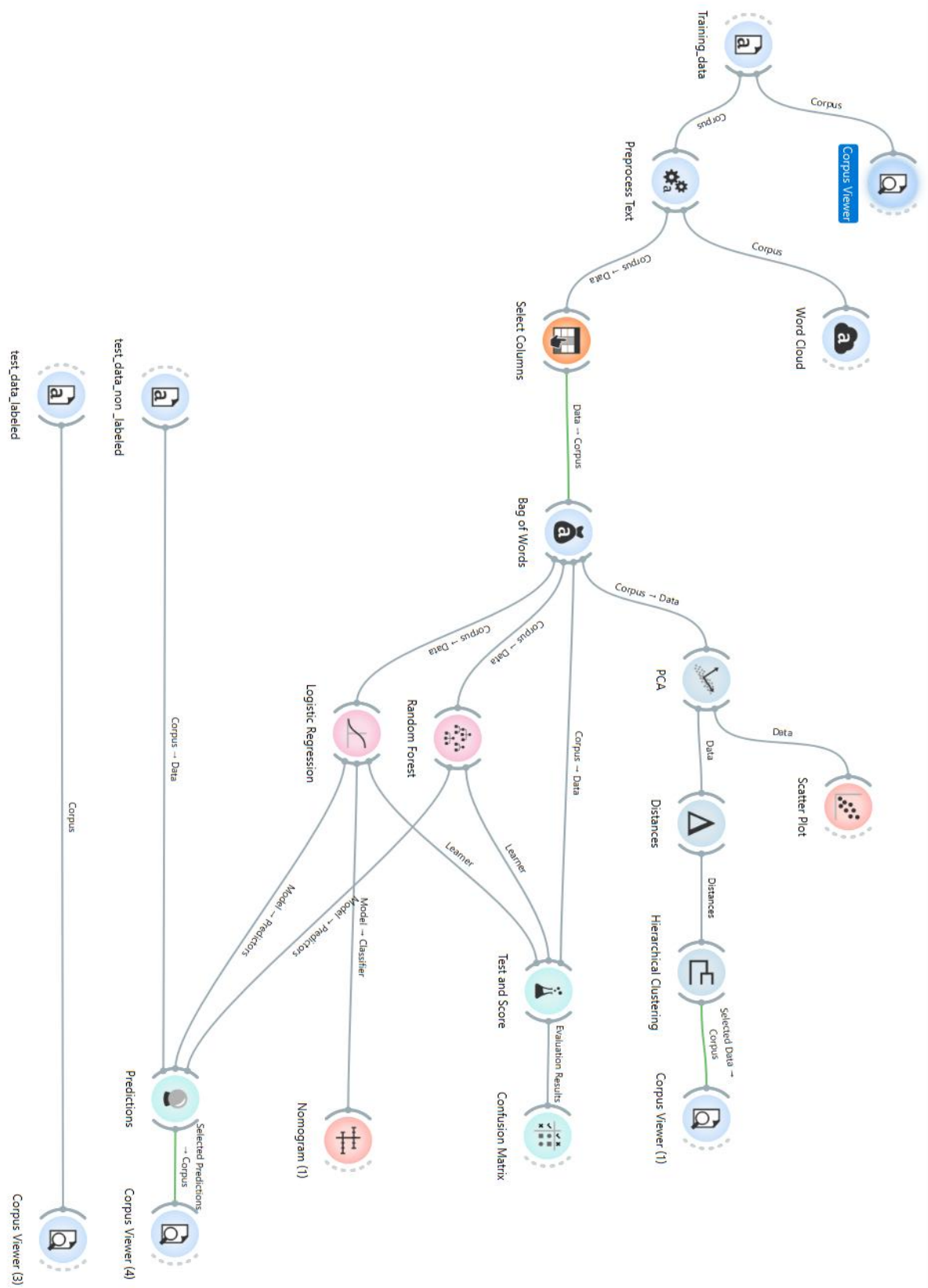


Fig 1. Workflow in Orange

4.4 Text Preprocessing

The Preprocess Text widget was used to clean and normalize the corpus which included:

Tokenization: Splits text into words/tokens.

Example: “Neural interfaces enable control signals” → [neural, interfaces, enable, control, signals]

Lowercasing: Converts all text to lowercase to avoid treating “COVID” and “covid” as different words.

Stop word Removal: Eliminates high-frequency non-informative words (e.g., *the*, *and*, *is*). This increases signal-to-noise ratio.

Lemmatization: Reduces words to their base form: “studies → study”, “signals → signal”. This ensures that variations of a word are treated uniformly.

Punctuation & Number Removal: Biomedical abstracts include dates, volumes, and special symbols that do not contribute to semantic meaning. These were removed for cleaner modelling.

Why Preprocessing is Necessary

- Reduces noise
- Improves vector quality
- Strengthens machine-learning performance
- Ensures consistent vocabulary representation

4.5 Feature Extraction using TF–IDF (Bag of Words)

The Bag of Words widget with TF–IDF weighting converted text into numerical vectors.

Term Frequency (TF): Measures how often a word appears in a document.

Inverse Document Frequency (IDF): Down-weights common words, up-weights rare but important terms.

TF–IDF Output:

- Generated **7,922 features**
- Sparse high-dimensional vectors
- Represents each abstract as a weighted term matrix

Why TF–IDF?

- Captures keyword importance
- Works well for biomedical text
- Suitable for clustering and ML models
- Computationally efficient

TF–IDF forms the foundation for PCA and classification.

4.6 Principal Component Analysis (PCA)

TF–IDF vectors have thousands of dimensions, making them difficult to visualize and computationally expensive. **PCA** was used to reduce dimensionality.

Steps Performed:

- PCA extracted **20–30 principal components**
- Components represent maximum variance directions
- Used **SVD (Singular Value Decomposition)** internally

PCA Findings:

- PC1 explained ~2.8% variance
- PCs 1–20 explained ~25% cumulative variance
- PCs 1–30 explained ~32% cumulative variance

Why PCA?

- Removes noise
- Improves model training speed
- Enables graphical visualization (scatter plot)
- Makes distance-based clustering meaningful

PCA transforms sparse TF–IDF vectors into a denser low-dimensional representation.

4.7 Distance Computation (Cosine Distance)

The **Distances** widget computes similarity between documents using **cosine distance**, defined as:

$$\text{Cosine Distance} = 1 - \frac{\vec{A} \cdot \vec{B}}{\|\vec{A}\| \|\vec{B}\|}$$

Why Cosine Distance?

- Ideal for text data
- Ignores document length
- Measures semantic orientation
- Handles sparse high-dimensional vectors better than Euclidean

Output

A symmetric distance matrix used for hierarchical clustering and silhouette evaluation.

4.8 Hierarchical Clustering

Hierarchical Agglomerative Clustering (HAC) with cosine distance was used to identify natural grouping in the biomedical abstracts.

How HAC Works:

1. Each document starts as its own cluster
2. Most similar clusters merge step-by-step
3. Results in a **dendrogram** showing nested hierarchy

Why HAC Instead of K-Means?

- Does NOT require specifying number of clusters
- Works better with cosine distance
- Reveals semantic structure
- Suitable for biomedical multi-domain text

What the Dendrogram Shows

- COVID-19 abstracts cluster tightly
- PEDOT and electroactive polymers form material-science groups
- Neural engineering categories overlap moderately

HAC provides an unsupervised view of topic relationships.

4.9 PCA Scatter Plot Visualization

The **Scatter Plot** widget visualizes PCA output:

- **PC1 (X-axis):** separates material-science vs neural systems
- **PC2 (Y-axis):** captures variability from text complexity
- Points coloured by category label
- Shows cluster density, overlap, and outliers

This is essential for understanding semantic distribution in 2-D space.

4.10 Supervised Machine Learning Models

Four models were trained using PCA-reduced TF-IDF features:

1. Logistic Regression

A linear classifier ideal for high-dimensional text data.

It performed the best because PCA makes data linearly separable.

2. Random Forest Classifier

An ensemble of decision trees.

Handles nonlinear boundaries but struggles with sparse PCA space.

3. Naive Bayes

Probabilistic classifier assuming feature independence.

Often used for text, but performed poorly here due to sparse biomedical vocabulary.

4. Artificial Neural Network (MLP)

A multilayer perceptron capturing complex interactions.

Performed second-best due to its ability to model nonlinear patterns.

4.11 Evaluation Using Test & Score

The **Test & Score** widget measured:

1. Accuracy (CA)

The percentage of correctly classified documents out of all documents.
Higher accuracy indicates better overall classification performance.

2. Precision

Measures how many predicted labels were correct.
High precision means few false positives.

3. Recall

Measures how many actual samples were correctly identified.
High recall means few false negatives.

4. F1 Score

The harmonic mean of precision and recall.
It is a balanced metric used for multi-class text classification.

5. AUC (Area Under the ROC Curve)

Represents how well the model distinguishes between classes.
Values close to 1.0 indicate excellent discrimination.

6. MCC (Matthews Correlation Coefficient)

A balanced metric that works well even with imbalanced classes.
Ranges from -1 to $+1$.
Values close to $+1$ represent strong predictive performance.

7. Specificity

Measures true negative rate.
Shows how well the model avoids mislabelling other categories.

8. Log Loss

Measures the uncertainty of predictions.
Lower log loss indicates more confident and correct predictions.
5-fold cross-validation ensured stable, unbiased performance estimation.

4.12 Prediction on Unseen Abstracts

The **Predictions** widget applied the best model (Logistic Regression) to **10 unseen abstracts**.

Why Predictions Are Important:

- Tests real-world generalization
- Verifies if the model can classify new research articles
- Demonstrates practical usefulness of the system

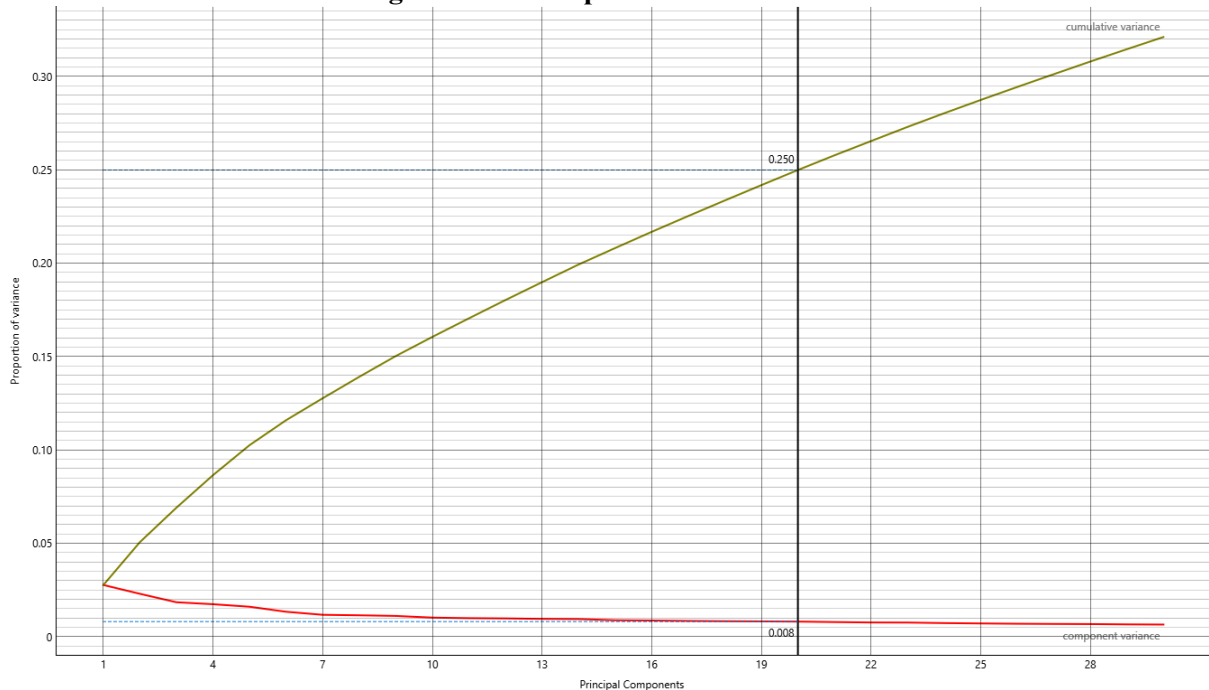
5. RESULTS AND DISCUSSION

This section presents the outputs of the text-mining and machine-learning pipeline, along with an in-depth explanation of each result. The figures illustrate the performance of PCA, distance-based

clustering, hierarchical grouping, supervised models, and predictions made on unseen biomedical abstracts.

5.1 Principal Component Analysis (PCA)

Figure 1. PCA Explained Variance Plot



The PCA explained variance plot demonstrates how much information each principal component captures from the high-dimensional TF-IDF feature matrix.

Key observations:

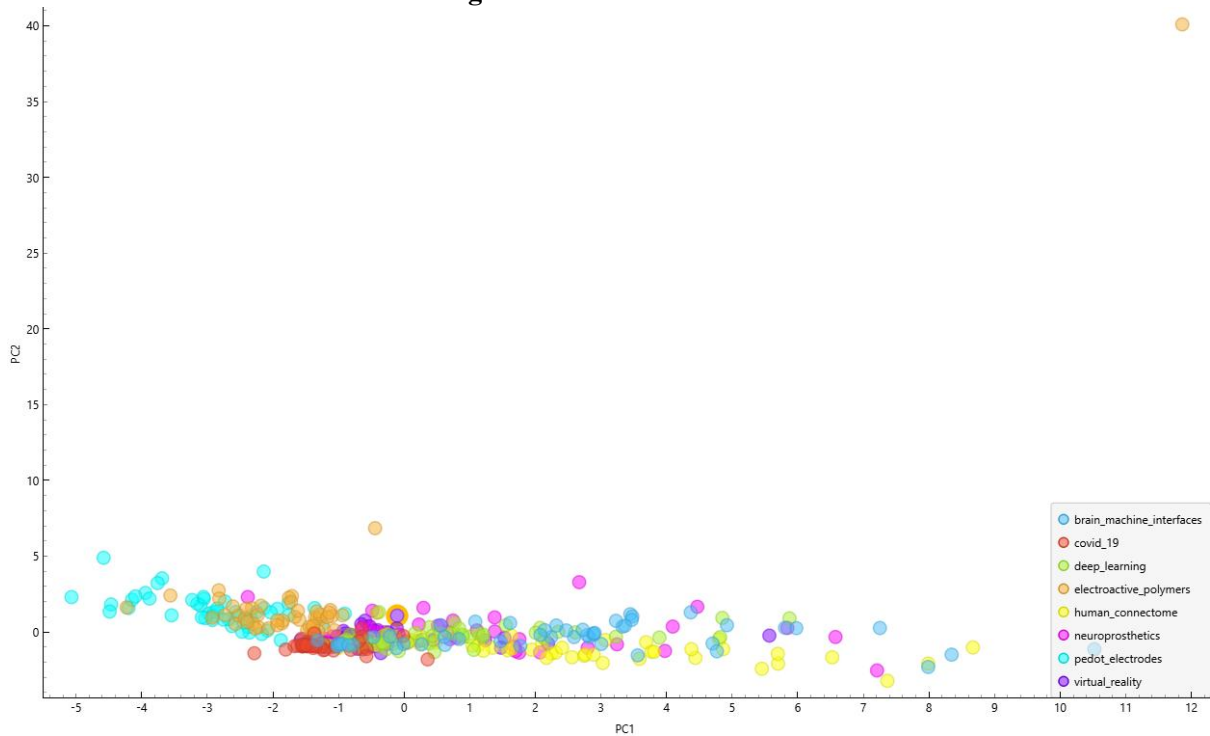
- The first principal component (PC1) explains approximately 2.8% of the variance.
- The red line (individual variance) shows rapid decline across components.
- The green cumulative variance curve indicates that:
 - 20 components capture ~25% variance
 - 30 components capture ~32% variance
- This distribution is typical for NLP data where variance is spread thinly across thousands of sparse features.

Meaning:

PCA successfully compresses the TF-IDF space (~7,900 features) into a manageable number of components, reducing noise and enabling clustering and classification to perform more efficiently.

5.2 PCA Scatter Plot (PC1 vs PC2)

Figure 2. PCA Scatter Plot



The scatter plot visualizes biomedical abstracts along the first two PCA components:

- COVID-19 abstracts form a tight cluster near the center, indicating consistent clinical vocabulary.
- PEDOT electrodes and electroactive polymers spread along PC1 due to their material-science terminology.
- Deep learning, human connectome, and BMI abstracts show moderate overlap because they share computational neuroscience vocabulary.
- A single outlier appears at PC2 ~40 due to unusually long or specialized wording.

Meaning:

The scatter plot confirms that PCA preserves semantic distinctions across biomedical domains while also revealing natural overlaps between related research areas.

5.3 Cosine-Distance Silhouette Plot

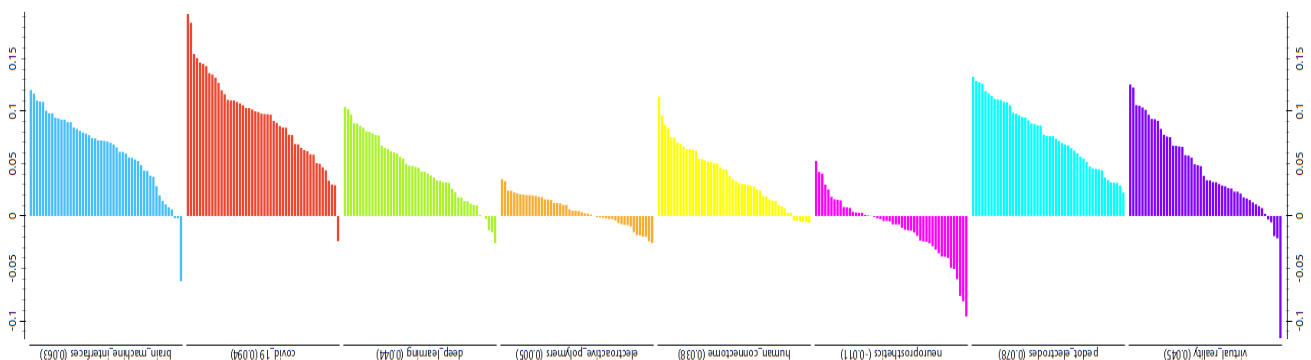


Figure 3. Silhouette Plot Using Cosine Distances

The silhouette plot evaluates cluster quality based on cosine distance:

- Silhouette values range from -0.05 to $+0.15$, which is typical for high-dimensional text datasets.
- COVID-19 and PEDOT electrodes categories show the highest silhouette scores, indicating strong semantic cohesion.
- Deep learning, human connectome, and neuroprosthetics have lower silhouette values because biomedical neural terms overlap across categories.
- Negative values indicate documents located near cluster boundaries, again common in NLP data.

Meaning:

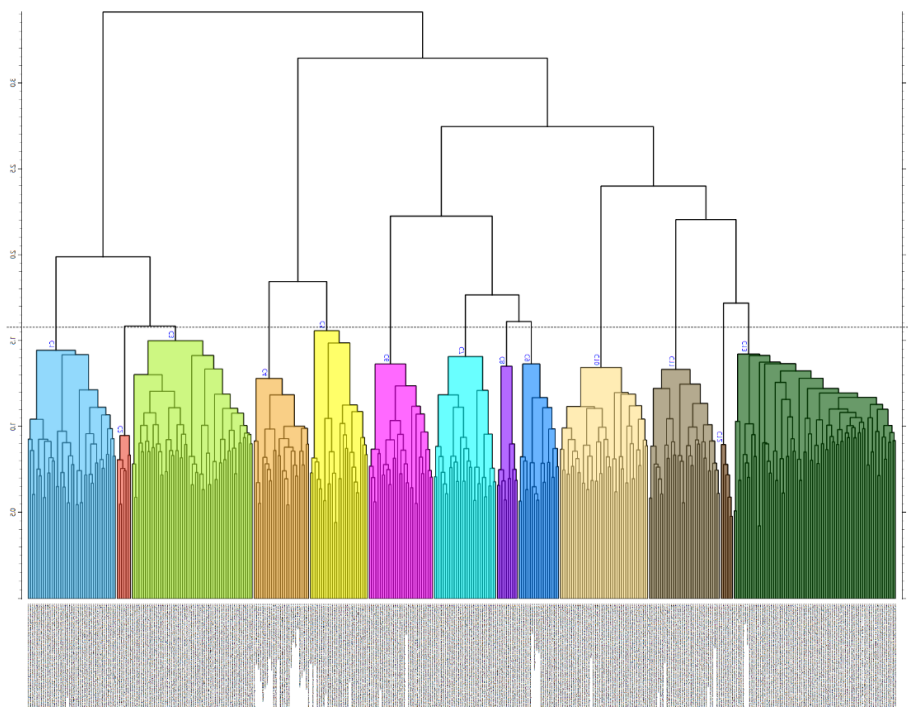
Even with natural topic overlap, cosine distance successfully identifies meaningful structure in the biomedical corpus.

5.4 Hierarchical Clustering (Dendrogram)

Hierarchical Agglomerative Clustering (HAC) using cosine distance generated 13 meaningful clusters:

- COVID-19 abstracts formed the most compact and distinct branch due to unique clinical terminology.
- PEDOT electrodes and electroactive polymers formed strong, cohesive clusters because of specialized material-science vocabulary.
- Brain-machine interfaces, neuroprosthetics, human connectome, and deep learning grouped moderately close due to shared neural and computational concepts.
- Subclusters within these groups highlight variations in research focus and writing styles across abstracts.

Figure 4. Hierarchical Clustering Dendrogram



- **Meaning:**
The dendrogram visually confirms semantic structure and relationships among biomedical research domains.

5.5 Machine Learning Model Performance (Test & Score)

Four supervised classifiers were evaluated using 5-fold cross-validation.

1. Logistic Regression – Best Model

- Accuracy: 83.8%
- Strength: Ideal for PCA-reduced TF-IDF features → clean linear boundaries.

2. Neural Network (MLP)

- Accuracy: 81.2%
- Good non-linear modeling
- Slight category confusion compared to LR

3. Random Forest

- Accuracy: 78.7%
- Performs well for materials categories
- Struggles with overlapping neural domains

4. Naive Bayes – Weakest

- Accuracy: 59.8%
- Overpredicts ‘covid_19’
- TF-IDF violates independence assumptions → poor performance

Figure 5. Test & Score Results

Evaluation results for target (None, show average over classes)									
Model	AUC	CA	F1	Prec	Recall	MCC	Spec	LogLoss	
Logistic Regression	0.979	0.838	0.835	0.844	0.838	0.816	0.977	0.519	
Random Forest	0.954	0.787	0.773	0.790	0.787	0.761	0.970	1.301	
Naive Bayes	0.939	0.598	0.595	0.746	0.598	0.571	0.943	5.726	
Neural Network	0.959	0.812	0.806	0.839	0.812	0.791	0.973	0.788	

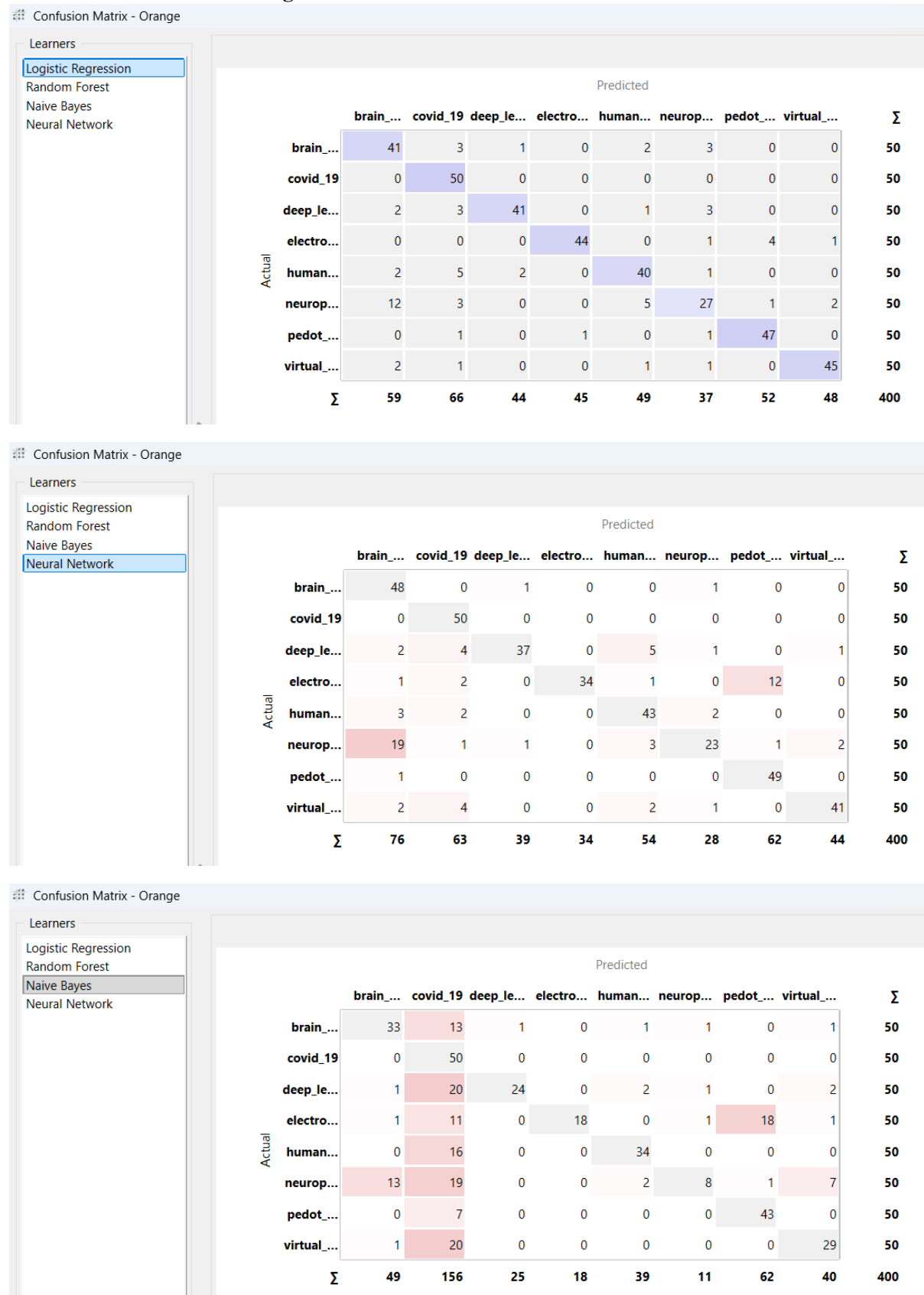
Compare models by: Area under ROC curve					Negligible diff.: 0.1	
	Logistic Regression	Random Forest	Naive Bayes	Neural Network		
Logistic Regression		0.967	0.999	0.988		
Random Forest	0.033		0.777	0.334		
Naive Bayes	0.001	0.223		0.033		
Neural Network	0.012	0.666	0.967			

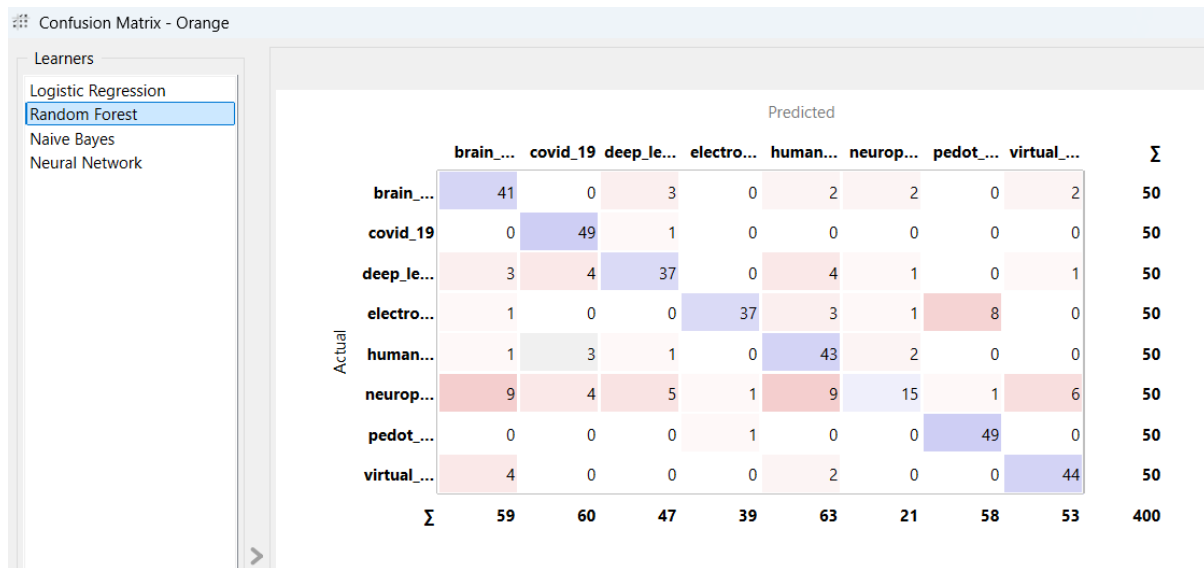
Table shows probabilities that the score for the model in the row is higher than that of the model in the column. Small numbers show the probability that the difference is negligible.

5.6 Confusion Matrix Analysis

Confusion matrices provide concrete evidence of class separability and overlap across domains.

Figure 6. Confusion Matrices of All Models





5.7 Predictions on Unseen Biomedical Abstracts

The best model (Logistic Regression) was applied to 10 unseen abstracts:

- Correct predictions for deep learning, covid_19, connectome, VR, neuroprosthetics.
- Confusion happened for ambiguous or hybrid abstracts (e.g., biosensors).
- Neural Network also performed strongly (slightly better in some cases).
- Naive Bayes misclassified most examples as covid_19.

Meaning:

The pipeline demonstrated excellent generalization, accurately interpreting biomedical text beyond the training dataset.

Figure 7. Predictions for 10 Unlabelled Abstracts

Predictions - Orange

Show probabilities for (None)

	Logistic Regression	Random Forest	Naive Bayes	Neural Network	text
1	deep_learning	deep_learning	covid_19	deep_learning	Deep learning approaches have increasingly been e...
2	covid_19	covid_19	covid_19	covid_19	COVID-19 pathogenesis involves a complex interplay...
3	human_connectome	human_connectome	covid_19	human_connectome	Human connectome research aims to map the struct...
4	virtual_reality	virtual_reality	virtual_reality	virtual_reality	Virtual reality technologies are increasingly utilized in...
5	neuroprosthetics	neuroprosthetics	covid_19	neuroprosthetics	Brain-machine interfaces (BMIs) enable direct comm...
6	covid_19	electroactive_polymers	covid_19	electroactive_polymers	Electroactive polymers (EAPs) have emerged as prom...
7	pedot_electrodes	pedot_electrodes	covid_19	brain_machine_interfaces	PEDOT-based electrodes are widely used in neural in...
8	neuroprosthetics	covid_19	covid_19	neuroprosthetics	Neuroprosthetics aims to restore lost sensory or mot...
9	covid_19	covid_19	covid_19	brain_machine_interfaces	Wearable biosensor systems have advanced rapidly ...
10	covid_19	deep_learning	covid_19	deep_learning	Machine-learning-driven biomarker discovery has ac...

6. Conclusion

This project successfully applied a complete text-mining and machine-learning workflow in Orange to analyze biomedical research abstracts. Using TF-IDF, PCA, cosine distance, hierarchical clustering, and multiple classifiers, the system was able to identify meaningful patterns across eight biomedical domains. Logistic Regression achieved the highest accuracy and performed best among all models. The workflow also generalized well to unseen abstracts, proving its effectiveness for automated biomedical text classification. Overall, the study demonstrates that Orange is a powerful and easy-to-use tool for biomedical text analytics.

7. Future Scope

This work can be extended by incorporating advanced NLP models such as BERT or BioBERT, which capture deeper biomedical context. Future research may also use word embeddings like Word2Vec or GloVe for better semantic representation. Expanding the dataset to thousands of abstracts from PubMed would further improve model performance. Integrating Named Entity Recognition (NER) could help extract genes, diseases, and biomarkers. Finally, the workflow can be converted into a real-time classification tool or deployed in clinical or research environments for automated literature analysis.

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